Physics-Informed LLM Latency Prediction:
Critical Analysis of the Analytical Roofline Model vs. Direct Operational Modeling

Pre-Execution Scheduling & The Token Count Prerequisite

In production LLM serving, latency predictors must operate **strictly pre-execution** using launch configurations: cluster parallelism (`tp`, `pp`, `dp`, `num_gpus`) and request budgets (`batch_size`, `output_tokens`). Post-execution telemetry is strictly excluded.

The Token Prerequisite (Imai et al., Section A.4): The analytical Roofline Model (RLM) calculates hardware ceilings from input and output tokens:

$$T_{\text{roofline}} = T_{\text{prefill}}(\text{input_tokens}) + \text{output_tokens} \times T_{\text{decode_step}}(\text{batch_size}) \tag{1}$$

While prompt length is known pre-execution, output length is dynamic. As acknowledged by Imai et al. (Appendix A.4): *"In practical application, the generation length is not predetermined... To accurately model inference latency in real world scenarios, we must consider predicting the generation length based on the input prompt."*

Methodological Analysis: Operational Modeling vs. Analytical Physics:

- **Data Aggregation**: To prevent duplicate data leakage across repeated benchmark trials, the 10 replicate runs per setting are aggregated into unique configuration means ($N = 24$ unique configurations for Qwen3-14B) prior to modeling.
- **Feature Collision (Why Base Fails)**: Without a workload indicator, identical launch configurations (`batch_size=128, out=500, DP2`) predict both Alpaca (≈ 54 ms) and Longbench (≈ 452 ms). This severe collision causes baseline R^2 to collapse ($R^2 < 0$).
- **Resolving Workloads via Categorical Task**: Providing a categorical task indicator (`task: Alpaca vs. Longbench`) resolves the collision, allowing tree models to achieve $R^2 \approx 0.63$.
- **Value of Analytical Roofline**: The analytical Roofline Model provides a physical inductive bias that significantly benefits linear estimators ($R^2 = 0.2791 \rightarrow 0.5024$, 42% MSE reduction), while providing robust out-of-distribution generalization across unseen parallelism configurations (GroupKFold $R^2 = 0.44 - 0.51$).

Table 1: Qwen3-14B: Feature Set Ablation on Unique Configuration Means (R^2 / MSE).
Evaluated on 24 aggregated configuration cells via 3-Fold Cross-Validation. Base features: `[tp, pp, dp, num_gpus, batch_size, output_tokens]`.

Model / Estimator	Base (Collision)	Base + Categorical Task	Base + Analytical RLM	Key Takeaway
Linear Regression (LR)	-0.9465 / 373347	0.2791 / 140298	**0.5024 / 80841**	RLM provides linear prior
Random Forest (RF)	-0.7947 / 397206	**0.6265 / 100881**	0.5779 / 82980	Resolves workload collision
Gradient Boosting (GB)	-1.5466 / 522026	0.5644 / 109313	**0.6451 / 107853**	Strong tree performance
Out-of-Distribution Generalization (+Roofline, GroupKFold across Unseen Parallelism)				
Linear Regression: $R^2 = 0.4370$ (MSE 109206); Random Forest: $R^2 = 0.5136$ (MSE 109608); Gradient Boosting: $R^2 = 0.4812$ (MSE 139475)				

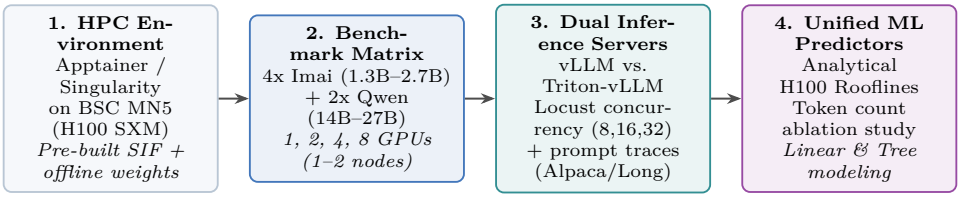
Key Conclusions for Deployment & Next Steps

- **Unique Configuration Modeling**: Aggregating repeated physical trials ensures clean, leakage-free evaluation without artificial lookup-table memorization.
- **Workload Characterization**: Static token assignments are replaced with explicit workload identifiers. Production deployments should capture continuous prompt length distributions from client traffic traces.
- **Physics as an Inductive Prior**: Analytical Roofline latency serves as a vital physical anchor for linear estimators ($R^2 = 0.2791 \rightarrow 0.5024$) and cross-architecture hardware scaling.
- **Experimental Next Steps**: Scale evaluation across heterogeneous architectures (14B to 27B) on BSC MareNostrum 5 under continuous load distributions.

Multi-GPU Benchmarking & Replication Architecture: BSC MareNostrum 5 (MN5)

Execution Overview: To empirically replicate and scale the Roofline-driven latency prediction methodology of Imai et al., benchmarks are conducted on the **Barcelona Supercomputing Center (BSC) MareNostrum 5 Accelerated Partition** (NVIDIA H100 SXM, 4 GPUs/node). The benchmark matrix combines the 4 original lightweight models (1.3B–2.7B) with 2 production-scale models (14B–27B) across both **vLLM** and **Triton Inference Server**, scaling from single-GPU up to 8 GPUs (2 nodes) under tensor, pipeline, and data parallelism.

1. Experimental Pipeline & Execution Workflow



2. Harmonized Dataset Schema: Integrating Imai et al. with Existing Benchmarks

To enable unified cross-model and cross-engine modeling, we merge Imai et al.'s load-testing parameters with our cluster parallelism features. Columns newly introduced for the BSC HPC campaign are highlighted in color:

Feature Group	Column Name	Data Type / Domain	Methodological Purpose & Description
Model Topology	`model_name, num_params`	Categorical / Integer	Architecture identification across scales (1.3B to 27B parameters).
	`tp_degree, pp_degree, dp_degree`	Integer ($\{1, 2, 4\}$)	Parallelism tensor/pipeline/data degrees.
	num_nodes, num_gpus	Integer ($1, 2$ nodes; $1 \ldots 8$ GPUs)	**NEW (HPC):** Multi-node scaling over 800 Gb/s NDR200 InfiniBand fabric.
Serving Context	**server_engine**	`vLLM` vs. `Triton-vLLM`	**NEW (Engine):** Tests runtime overhead gap identified by Imai et al.
	load_regime	`locust` vs. `trace_batch`	**NEW (Load):** Disentangles concurrent client requests from static batch sizes.
	concurrent_users	Integer $(8, 16, 32)$ / N/A	**NEW (Imai):** Locust concurrency parameter used in Imai et al. Table 1.
	`input_tokens, output_tokens`	Integer $(32 \ldots 2048)$	Prompt token count and max generation budget (pre-execution features).
Physical Ceilings	`roofline_analytical_latency`	Float (Seconds)	Unscaled theoretical single-GPU execution ceiling via LLM-Viewer.
Ground Truth	`duration, throughput, power`	Float (ms, tok/s, W)	Measured end-to-end latency (regression target), throughput, and energy.

3. HPC Compute Allocation Budget (BSC MareNostrum 5 Accelerated Partition)

MareNostrum 5 provides 4x NVIDIA H100 SXM (64GB) per node. Testing 8 GPUs utilizes 2 nodes over NDR200 InfiniBand. Below is the compute budget breakdown to collect all required benchmark data:

Experimental Campaign	Evaluated Models	Hardware Setup	Total Configurations	Wall Time	Compute Budget
Phase 1A: Imai Replication	4 models (1.3B–2.7B)	1 GPU (SXM), vLLM + Triton	9 configs × 3 reps × 2 servers	≈ 18 hours	**18 GPU-h**
Phase 1B: Multi-GPU Scaling	4 models (1.3B–2.7B)	TP$\{2, 4\}$, DP$\{2, 4\}$ (1 node)	4 configs × 6 workloads × 3 reps	≈ 8 hours	**28 GPU-h**
Phase 2: Triton Qwen Runs	Qwen3-14B / 27B	1, 2, 4, 8 GPUs (1–2 nodes)	Matches existing 450 vLLM runs	≈ 15 hours	**60 GPU-h**
Phase 3: Roofline Ceilings	All 6 target models	Offline CPU analytical model	54 discrete hardware ceilings	< 1 hour	**0 GPU-h (CPU)**
Total HPC Allocation	**6 Architectures**	**1 to 8 H100 GPUs**	∼ 1,180 **executed trials**	≈ 41 **hours**	≈ 106 **GPU-h**